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Descriptive Statistics
Academic Year 2025-2026
(Time: 3 hours)

Courses Questions: (05 points)

1- What are the principal methods of measuring trend from a given time series?
2- Name two discrete distributions and two continuous distributions
3- What are the different methods of random sampling
4- What is the difference between a univariate statistical series and a bivariate statistical series
5- What is an Index Number? What are the uses of index numbers in decision making?

Exercise II: (05 points)

Calculate the arithmetic, geometric, harmonic, and quadratic means for the following distribution: 8, 9, 12, 13, 15.

Exercise III: (05 points)

Three items X, Y, and Z were selected for the calculation of the cost-of-living index in Bamenda. We have the following information on prices (in FCFA/kg) and quantities consumed in (Kg) for the Year 2023 and 2024.

Items	P_0 2023 Q_0		P_1 2024 Q_1	
	Prices (in FCFA)	Quantity (in Kg)	Prices (in FCFA)	Quantity (in Kg)
X	13	7	16	5
Y	5	12	8	12
Z	15	9	13	10

From the following data compute Laspeyre's index number and Paasche's price index number

$$\text{Laspeyre's} = \sum \frac{P_1 Q_0}{P_0 Q_0} \times 100$$

$$\text{Paasche's} = \sum \frac{P_1 Q_1}{P_0 Q_1} \times 100$$

Exercise IV: (05 points)

In the following contingency table, X represents a country's trade balance and Y the exchange rate.

X \ Y	[0,10[	[10,20[	[20,30[
[0,4[	0	40	0
[4,8[	20	0	20

1- Determine the marginal distributions of X and Y

2- Calculate the means and variances of these marginals' distributions